

CSC 281
Discrete Structures
Quiz 1

Full Name: _____

Instructions:

- *Time: 70 Minutes*
- *This is a closed book, closed notes quiz.*
- *The last page of this booklet has intentionally been left blank to serve as scratch paper; however, it cannot be detached from the booklet. No other paper is allowed during the quiz.*
- *No cell phones, laptop, or tablet are on the desk during the quiz. Using a cell phone, laptop, or tablet for any reason during the quiz will result in a grade of 0. Any violations will result in a grade of F in the course and university disciplinary action.*
- *During the quiz, only the provided booklet is allowed on your desk. No other materials or items should be present.*
- *SHOW YOUR WORK. Answers with no work will be marked wrong regardless of the correctness of the final answer.*
- *Circle or otherwise indicate final answers as appropriate.*
- *The total points available for a problem are shown in parentheses at the beginning of each problem*
- *This quiz consists of 6 questions*

Section	Maximum Score	Your Score
Total	50	

1. (10) Let

p: "The device is on."

q: "The sensor is recording."

r: "The system is logging data."

Construct the truth table for the compound statement:

$$(\neg p \vee \neg q) \rightarrow \neg r$$

Then explain under what condition(s) the entire statement is false.

2. (10) Translate the following into symbolic form:

"There exists a student who has not taken Discrete Math."

Let $D(x)$: "x has taken Discrete Math."

Domain: All students at the university.

3. (10)

Let $A = \{2, 4, 6, 8\}$,

$B = \{4, 5, 6, 7\}$,

and $U = \{1, 2, \dots, 10\}$

Find:

(a) $A \cup B$

(b) $A \cap B$

(c) B' (complement of B with respect to U)

4. (10) Prove by contradiction: If $3n + 2$ is even, then n is even, where $n \in \mathbb{Z}$.

5. (10) The degrees of the vertices in a graph are: 2, 4, 2, 2, 4.
- (a) Does this graph have an Euler path or circuit?
 - (b) Briefly explain your answer using Euler's criteria.
6. For each of the following, provide a clear and rigorous mathematical proof. Avoid simply listing equations; your reasoning must be precise and logically sound.
- (a) Prove the uniqueness of square roots of positive real numbers.
 - (b) Let $f(x)=2x+1$. Prove, using contraposition, that if $f(x)$ is not even, then x is not an integer.